Gang Liu (Leo)



AI Algorithm Engineer / Forward-Deployed Engineer (FDE) — LLM Systems & Edge AI

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Education

Shanghai University of Engineering Science **M.Eng. in Electronic Information (AI track)** Sep 2024 – Jun 2027 (expected)

GPA: 3.36 / 4.0 Advisor: Prof. Xiang Liu Thesis: Multi-scale image detection for illegal-construction recognition Honors: First-Class Graduate Academic Scholarship

Guangling College, Yangzhou University **B.Sc. in Information and Computing Science** Sep 2017 – Jun 2021

GPA: 3.23 / 4.0 Honors: Outstanding Undergraduate Thesis Award

Technical Skills

• **LLM Inference Optimization & Edge Deployment**: Practice with Qwen quantization, QLoRA, ONNX Runtime and TensorRT; exploration of single-GPU inference and Jetson deployment.

• **LLM / VLM & Agent Systems**: Qwen2 / Qwen2.5-VL multimodal fine-tuning, RAG (ChromaDB retrieval & rerank), Prompt Engineering, LangChain / LangGraph, multi-agent orchestration, MCP protocol, Function Calling.

• **Computer Vision & Deep Learning**: PyTorch, YOLOv8 / YOLOv11 (P2 pyramid, multi-scale fusion, decoupled head, instance segmentation), small-object detection, monocular ranging, OpenCV.

• **AI-Native Engineering**: Claude Code, Codex, Antigravity, Grok, MiniMax, GLM, Kimi Code; FastAPI / React full-stack delivery; multi-agent agile workflow for end-to-end system builds.

Experience & Core Projects

Government-Enterprise AI Agent & RAG System — Zhuofan (current internship) | **LLM Agent / RAG** Sep 2026 – Present

Backend Development Engineer (AI Application Development Intern) [Stack: AI Agent / RAG / FastAPI / Function Calling]

• Contributed to core Agent logic: ReAct + workflow orchestration for intent recognition, task routing and result fusion; chained retrieval, pre-screening and business APIs via Tool Use / Function Calling.
• Built the policy RAG pipeline (ChromaDB + BM25 + Rerank); wrapped streaming output and session management in FastAPI (SSE) with timeout-retry, JSON Schema validation and human-handoff fallback.

Streetlight Detection & Monocular Ranging — CETC 50th Research Institute | **Production CV System** Oct 2025 – Jul 2026

Algorithm R&D Intern [Stack: YOLOv8 / Monocular Ranging / Kalman Filter / Industrial PC]

• Worked on YOLOv8 detection for slender pole targets and monocular ranging using camera calibration and bounding-box localization.
• Added Kalman temporal filtering against vibration-induced false alarms; deployed and optimized on a vehicle-mounted industrial PC — live in production inspection fleets.

Multi-Agent Cybersecurity Threat-Analysis System | **National 3rd Prize · Team Captain** Jul 2025 – Oct 2025

Team Captain — full-stack architecture & LLM R&D [Stack: Qwen2-7B / INT4 Quantization / QLoRA / Multi-Agent / RAG]

• Worked on Qwen2-7B quantization and QLoRA fine-tuning for single-GPU offline security analysis.
• Designed an agent workflow for parsing, threat-intelligence retrieval, attack-chain reasoning and reporting.

Master's Thesis — UAV Illegal-Construction Detection with Jetson Edge Deployment | **First Author / Lead** Oct 2024 – Present

Core of Master's research (Advisor: Prof. Xiang Liu) [Stack: YOLOv11s-seg / Jetson Orin Nano / TensorRT / Domain Adaptation]

• Built a UAV image dataset and explored high-resolution features and DySample for small-object segmentation with YOLOv11s-seg.
• Explored PyTorch → ONNX → TensorRT conversion and Jetson deployment, with video streaming and coordinate processing.

understand_mov_v2 — Multimodal Video Retrieval & Understanding Platform | **Solo Full-Stack Project** Apr 2026 – Jun 2026

Independent design & development [Stack: Multimodal Retrieval / FastAPI / React / SQLite FTS5]

• Built FFmpeg frame extraction, multimodal analysis and SQLite FTS5 text search, with a persistent task queue, recovery and retries; public demo is static.

Honors & Publications

• National Second Prize — China Mathematical Contest in Modeling (Huawei Cup), 2025 (Team Captain)

• National Third Prizes — Huawei Cup Post-Graduate Cybersecurity Competition 2025 & 7th Post-Graduate AI Innovation Competition 2025 (both as Team Captain); National IC EDA Elite Innovation Competition 2025 (Core Member)

• Provincial First Prize — CIMC Siemens Cup 2025; Provincial Second Prizes — CIMC 2026 & Post-Graduate EDC 2026

• Research — remote-sensing image detection and lightweight change detection; manuscript work in progress.

• Certification — CCF Large Model Competence Certification, Adult track (LMCC-A), Level 1 (2026-09-19, China Computer Federation).